App Dev-1 Project Report

1.Student Details

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About Me: I am a dual degree undergraduate student currently pursuing a BS in Data Science. My interests lie in backend web development and database-driven applications. Through this Trekking Management Application project, I developed a role-based web application using Flask, SQLAlchemy, and SQLite, gaining hands-on experience in database design, authentication, role-based access control, and the development of dashboard-driven management systems following the MVC architecture.

2.Project Details

Project Title: Trekking Management Application

Problem Statement:

The Trekking Management Application is a multi-user web application designed to manage trekking activities efficiently. The system connects three types of users: Admin, Trek Staff, and Trekkers. The application enables administrators to manage treks, assign staff, and monitor bookings, while trek staff can manage assigned treks and update trek statuses. Trekkers can browse available treks, make bookings, and view their booking history. The application streamlines trekking operations by centralizing trek management, staff coordination, and user bookings within a single system.

Approach:

I developed this project incrementally, beginning with database design and authentication, followed by the implementation of trek management, staff assignment, trek booking, booking history, and status tracking modules. I used Bootstrap to create a responsive and user-friendly interface, while Jinja2 templates enabled dynamic rendering of web pages. Throughout the development process, validation checks such as duplicate booking prevention, slot availability verification, and access restrictions were incorporated to ensure reliable system behavior.

3.AI / LLM Declaration

I used ChatGPT (GPT-5) for assistance during the development of this project. The assistance included understanding Flask concepts,Bootstrap,debugging suggestions, improving HTML/CSS layout, and formatting documentation.The extent of AI/LLM usage is approximately **30-35%**, mainly for guidance and code suggestions and debugging.

4.Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework
SQLAlchemy	Object Relational Mapper for SQLite database
SQLite	Local Database for storing application data
Jinja2	Template engine for rendering dynamic HTML Pages
HTML	Structure of application web pages
CSS	Styling and layout of the application
Bootstrap	Responsive UI Components and Dashboard design
Flask-Login	User authentication,session management and role based access control

5.Database Schema / ER Diagram

Tables

User → id (PK), username, full_name, email, phone, password, role, approval_status, is_active

StaffProfile → id (FK), phone, experience, status

Trek → id (PK), trek_name, destination, difficulty, duration, price, available_slots, start_date, end_date, description, status, assigned_staff_id (FK)

Booking → id, user_id (FK), trek_id (FK), booking_date, booking_status, payment_status

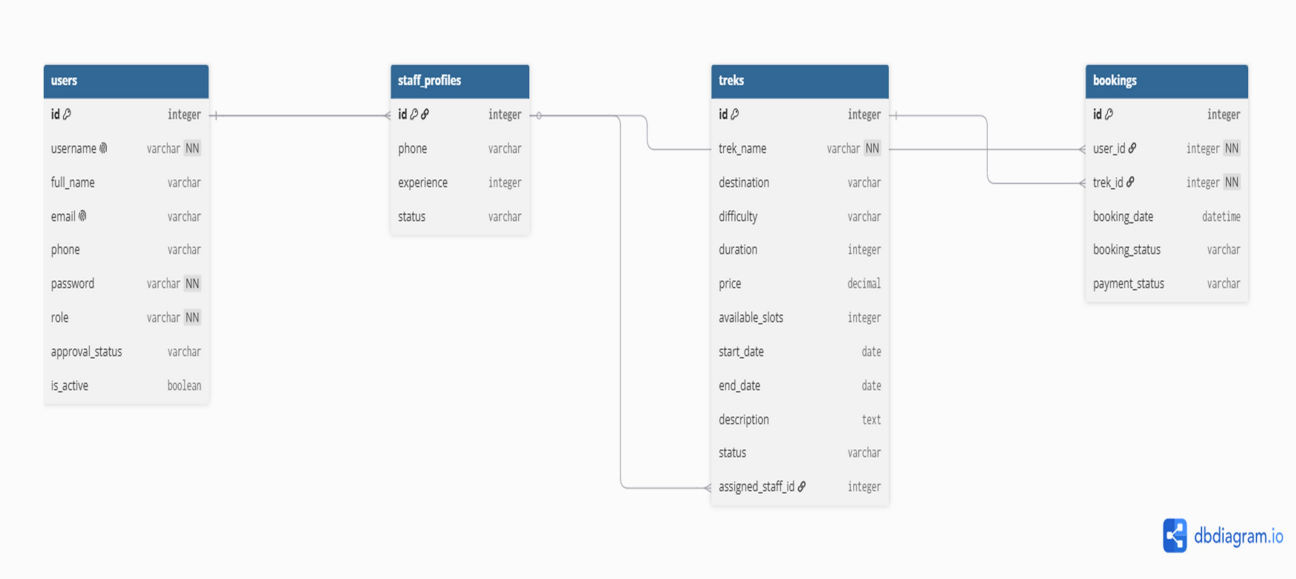
Relationships

User ↔ StaffProfile (One-to-One)

StaffProfile → Trek (One-to-Many)

User → Booking (One-to-Many)

Trek → Booking (One-to-Many)



6.Functionalities

Authentication

- Trekkers and Trek Staff can register and log in to the application.
- Admin account is pre-created and manages the overall system.
- Role-based authentication ensures users can access only their authorized modules.

Admin Features

- Approve or reject Trek Staff registrations.
- Create, update, and delete trek records.
- Assign staff members to treks.
- View and manage all treks, staff, trekkers, and bookings.
- Monitor trek availability and booking history.

Trek Staff Features

- Register and log in to the application.
- View assigned treks.
- Update trek status (Open, Closed, Ongoing, Completed).
- View participants registered for assigned treks.
- Update personal profile information.

Trekker Features

- Register and log in to the application.
- Browse available treks.
- Book available treks.
- View booking history and booking status.
- Update personal profile information.

Additional Features

- Prevention of duplicate trek bookings.
- Validation of available trek slots to prevent overbooking.
- Role-based access control using Flask-Login.
- Dashboard-based management for Admin, Trek Staff, and Trekkers.
- Booking history tracking for trekkers and administrators.

7.Video Presentation

Drive Link:
https://drive.google.com/file/d/1nEqWpt_Ox8tFMl-u5WGKSQpGY_UYeeom/view?usp=sharing